

Asynchronous communications (MQTT)

What is MQTT?

Message Queuing Telemetry Transport (MQTT)

A lightweight publish/subscribe messaging protocol designed for:

- constrained devices
- unreliable or low-bandwidth networks
- IoT and edge computing scenarios

👉 Unlike HTTP (request/response), MQTT is **event-driven and asynchronous**.

Key Characteristics

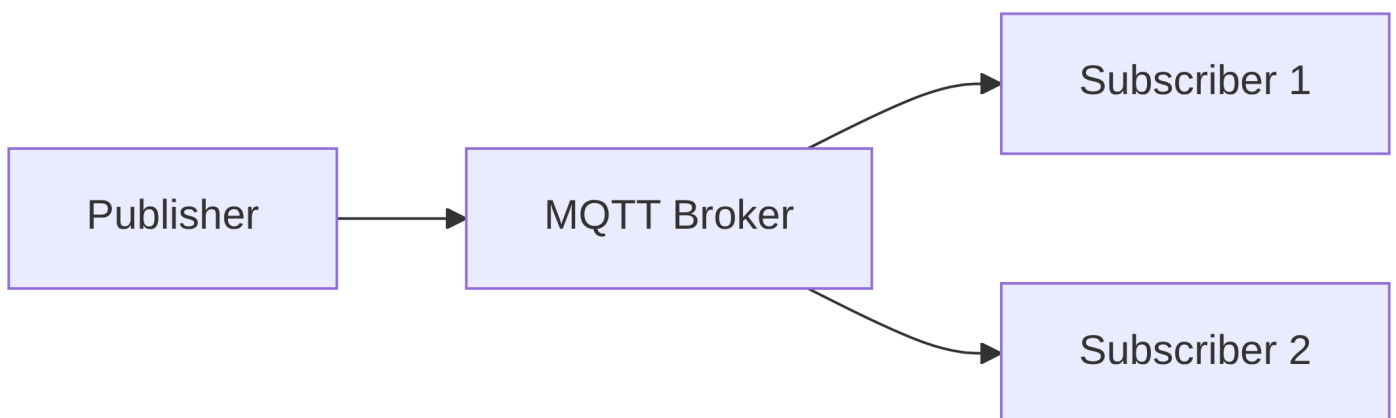
- Low bandwidth usage
- Low latency
- Asynchronous communication
- Decoupled architecture
- Native fit for IoT

👉 Ideal when you have many distributed producers and consumers.

MQTT Architecture

Core components:

- **Publisher** → sends messages
- **Broker** → routes messages
- **Subscriber** → receives messages



👉 The broker is the central component.

Decoupling

MQTT enables three types of decoupling:

- **Temporal** → clients do not need to be online at the same time
- **Spatial** → clients do not know each other
- **Technological** → different languages and platforms

👉 This is why MQTT works well with microservices.

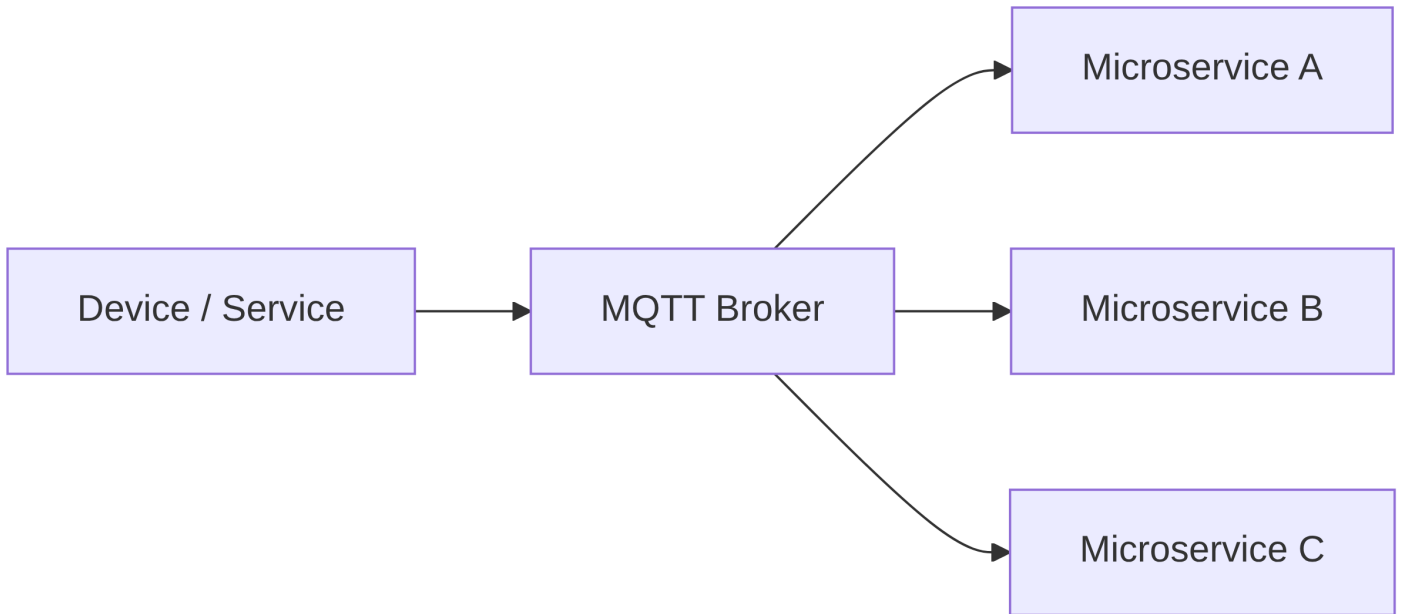
MQTT in Microservices

MQTT acts as a:

👉 Lightweight Event Bus

Typical use cases:

- notifications
- telemetry
- real-time updates



Brokers

Common MQTT brokers:

- Mosquitto → lightweight, open-source
- HiveMQ → enterprise-grade
- EMQX → high-performance, clustered
- RabbitMQ (MQTT plugin)
- Cloud solutions (AWS IoT, Azure IoT)

Message Structure

An MQTT message includes:

- **Topic** → routing key
- **Payload** → data (JSON, binary, etc.)
- **QoS** → delivery guarantee
- **Retain flag**
- **Properties** (MQTT 5.0)

Topics

Topics are hierarchical strings:

```
bank/account/123/deposit  
sensors/temperature/lab1
```

👉 Topics define routing, not the payload.

Topic Design (Best Practices)

Recommended pattern:

```
<domain>/<entity>/<id>/<event>
```

Examples:

- `iot/device/thermostat-1/temperature`
- `bank/account/123/withdraw`

! Avoid:

- generic topics like `data` , `events`
- encoding routing logic inside payloads

Subscriptions & Wildcards

- `+` → matches one level
- `#` → matches multiple levels

Examples:

```
home/+ /temperature  
home/#
```

Quality of Service (QoS)

- QoS 0 → at most once
- QoS 1 → at least once
- QoS 2 → exactly once

QoS Trade-offs

QoS	Guarantee	Cost	Typical Use
0	None	Low	High-frequency sensors
1	At least once	Medium	Business events
2	Exactly once	High	Critical operations

👉 QoS 2 is rarely used in practice due to overhead.

👉 QoS 1 may produce duplicates

- idempotent consumers
- message IDs
- deduplication logic

Retained Messages vs Event Messages (MQTT)

Retained Messages

- The broker stores the **last message per topic**
- New subscribers immediately receive the **latest retained value**

👉 This is used to represent the **current state** of a system

Typical use case:

- device status (online/offline)

- current configuration
 - latest known sensor value
-

Event Messages (Non-Retained)

- Messages are **transient**
- Delivered only at the time of publication
- Not stored by the broker

👉 This represents a **change or event over time**

Typical use case:

- temperature readings
 - alerts
 - activity logs
-

Key Insight

👉 **Retained messages represent “what is true now”**

👉 **Event messages represent “what just happened”**

Last Will and Testament (LWT)

- Defined when the client connects
- Published by the broker on unexpected disconnect



Use case:

- failure detection in distributed systems
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Persistent Sessions (MQTT)

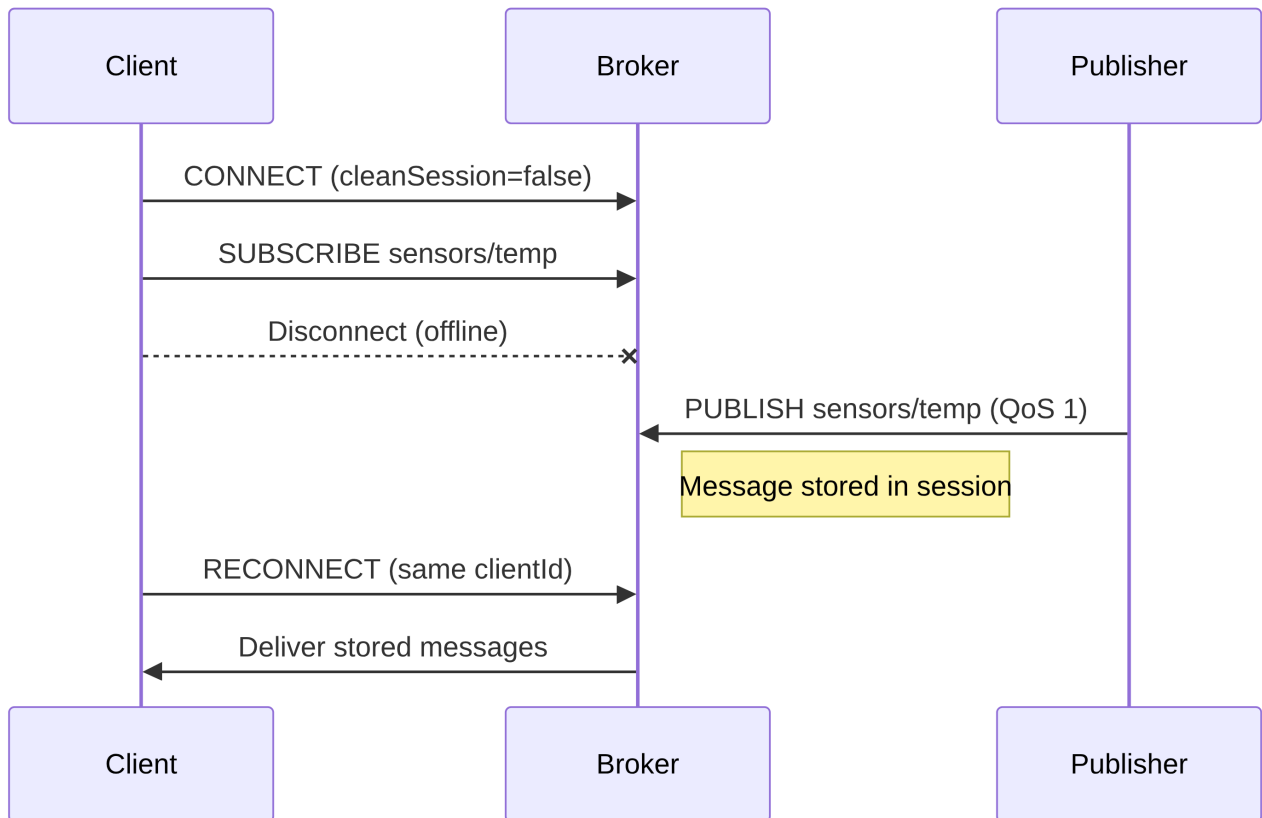
clean session = false

When a client reconnects, the broker preserves its session state instead of resetting everything.

The broker stores:

- subscriptions
- undelivered QoS 1/2 messages
- in-flight messages (awaiting acknowledgment)

👉 Useful for intermittently connected devices (IoT, mobile, edge systems)

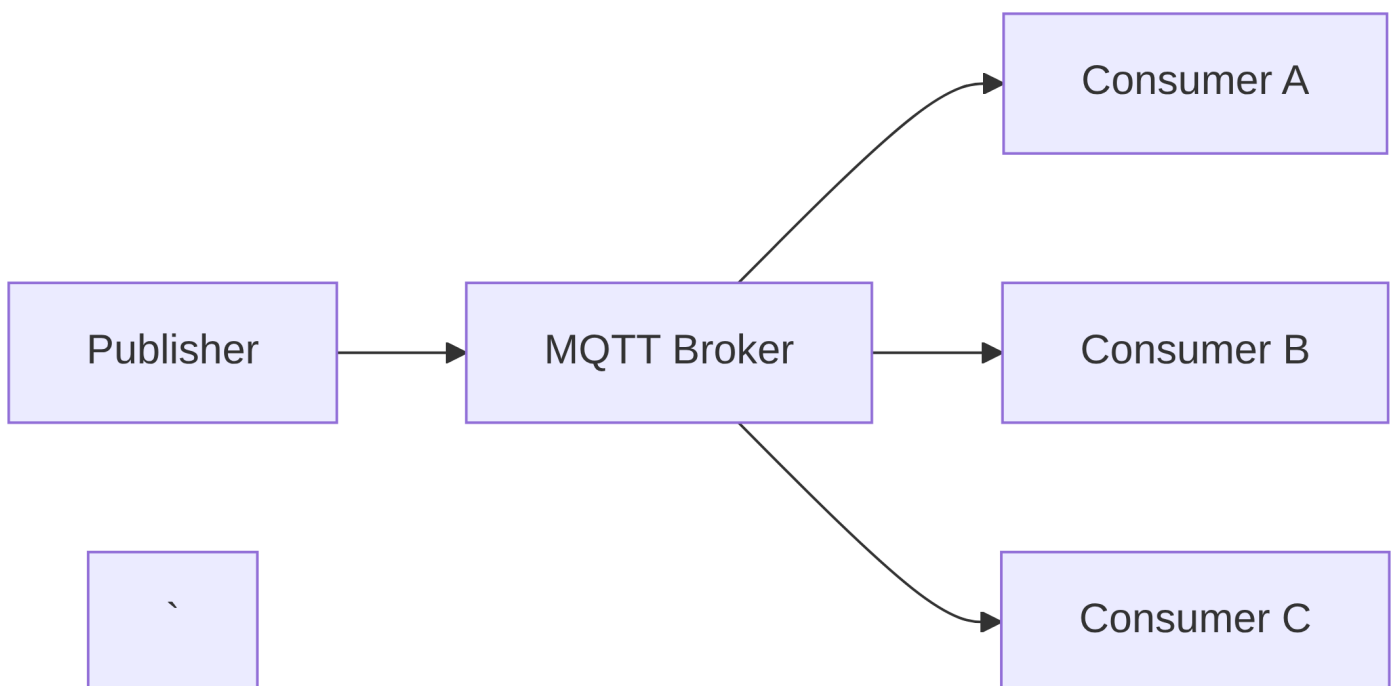


Multiple Consumers on the Same MQTT Topic

Default Behavior (Standard MQTT)

In MQTT, when multiple clients subscribe to the **same topic**, each of them receives a **full copy** of every message.

👉 This is a **publish/subscribe broadcast model (fan-out)**



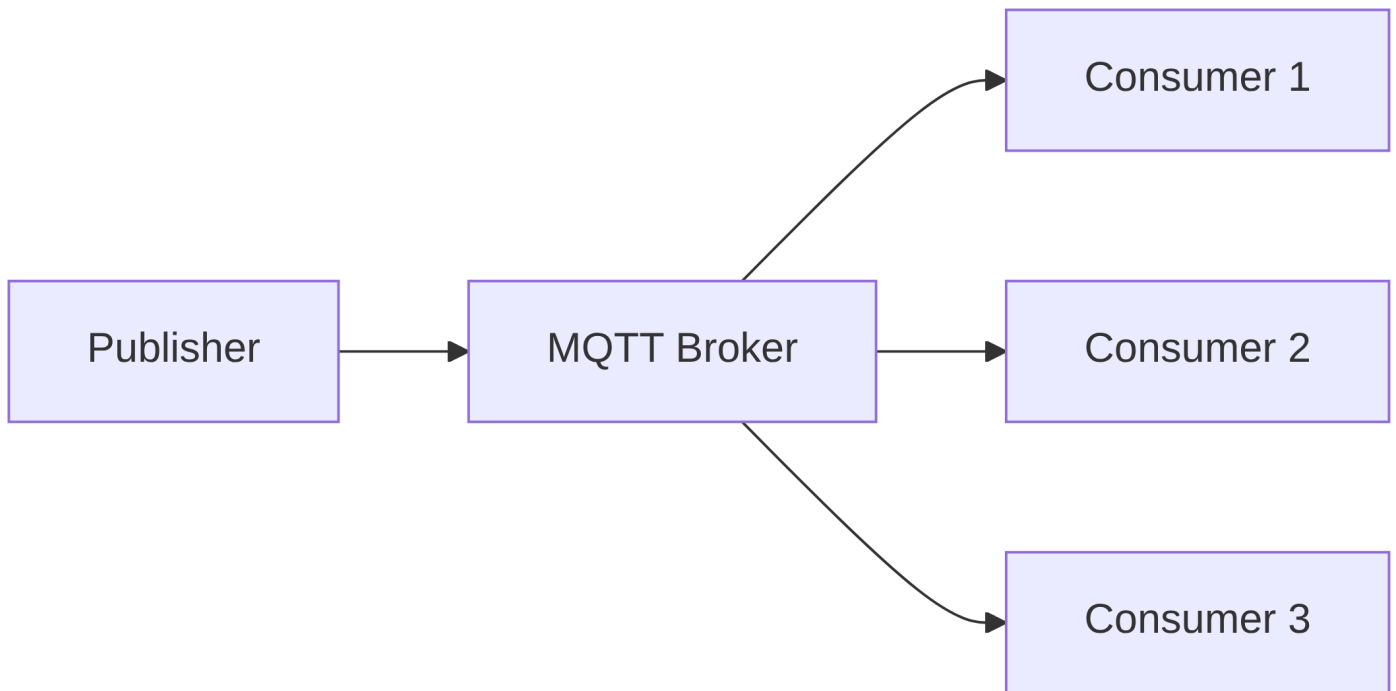
- Each subscriber receives **all messages**
- There is **no load balancing**

- Consumers are **independent of each other**

Shared Subscriptions (MQTT 5.0)

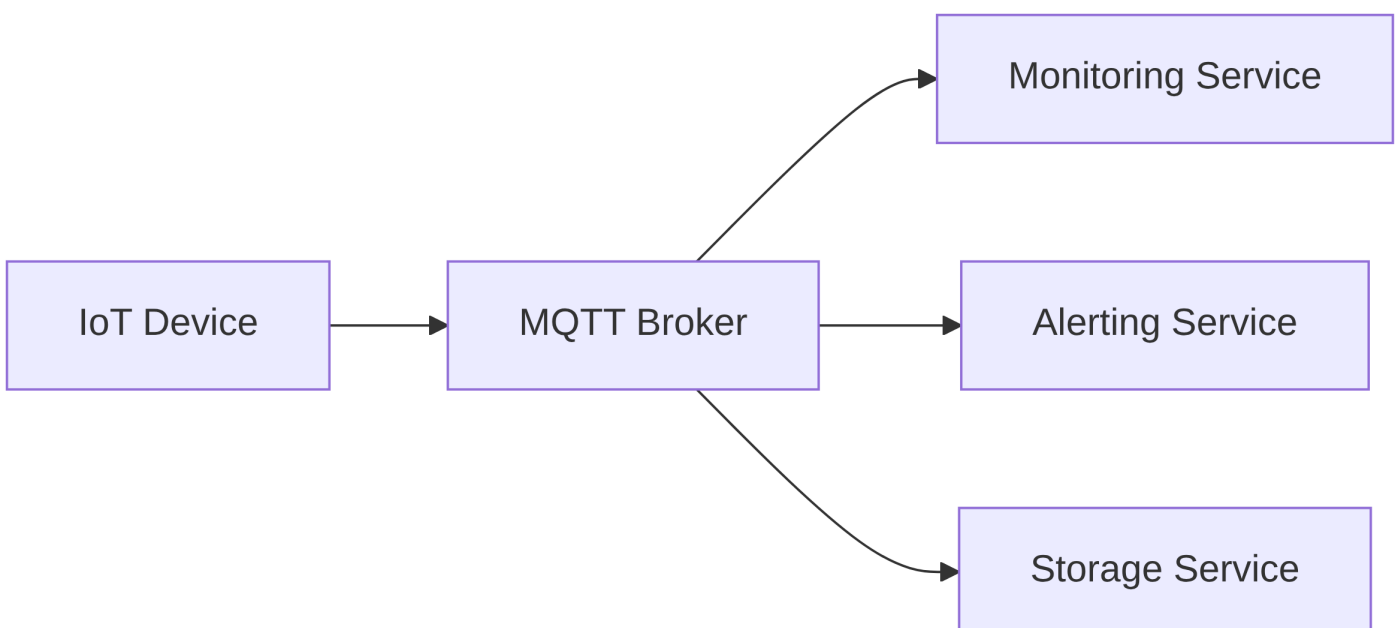
MQTT 5.0 introduces **shared subscriptions** for load balancing:

```
$share/group1/sensors/temp
```



- each message is delivered to **only one consumer in the group**
- enables **horizontal scaling**
- behaves similarly to a **queue / consumer group**

End-to-End Example



Flow:

1. Device publishes temperature
 2. Broker distributes message
 3. Multiple services react independently
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Integration Patterns

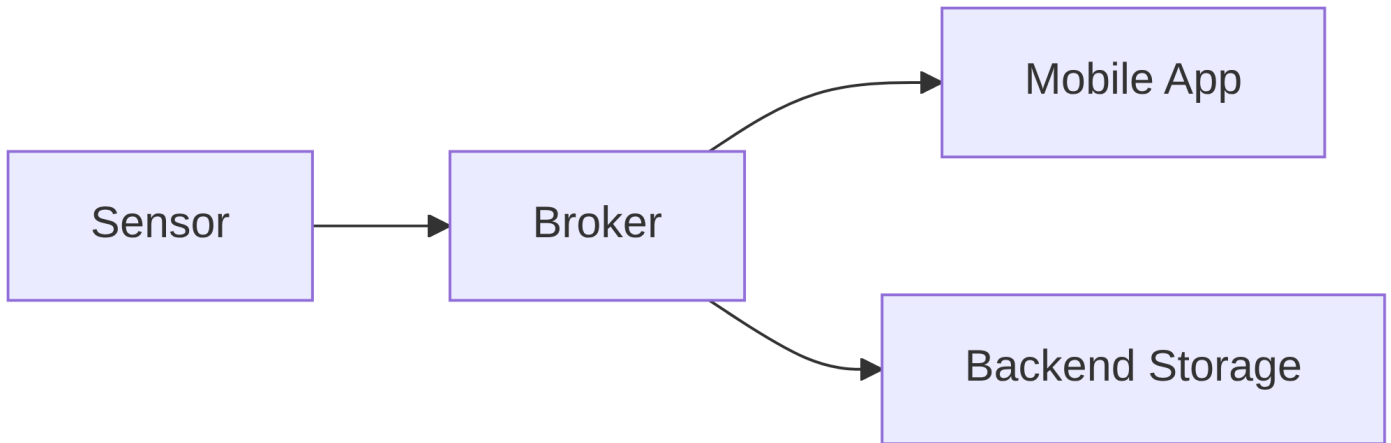
- MQTT → Kafka bridge
- MQTT → REST gateway
- MQTT → database sink

👉 MQTT is often part of a larger architecture.

Case Study 1: Smart Home

Scenario:

- Sensors publish temperature and humidity
- Mobile app subscribes for real-time updates
- Backend stores data



Key benefits:

- real-time updates
 - low power consumption
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Case Study 2: Industrial IoT

Scenario:

- Machines publish telemetry
- Monitoring service detects anomalies
- Alerting system triggers alarms



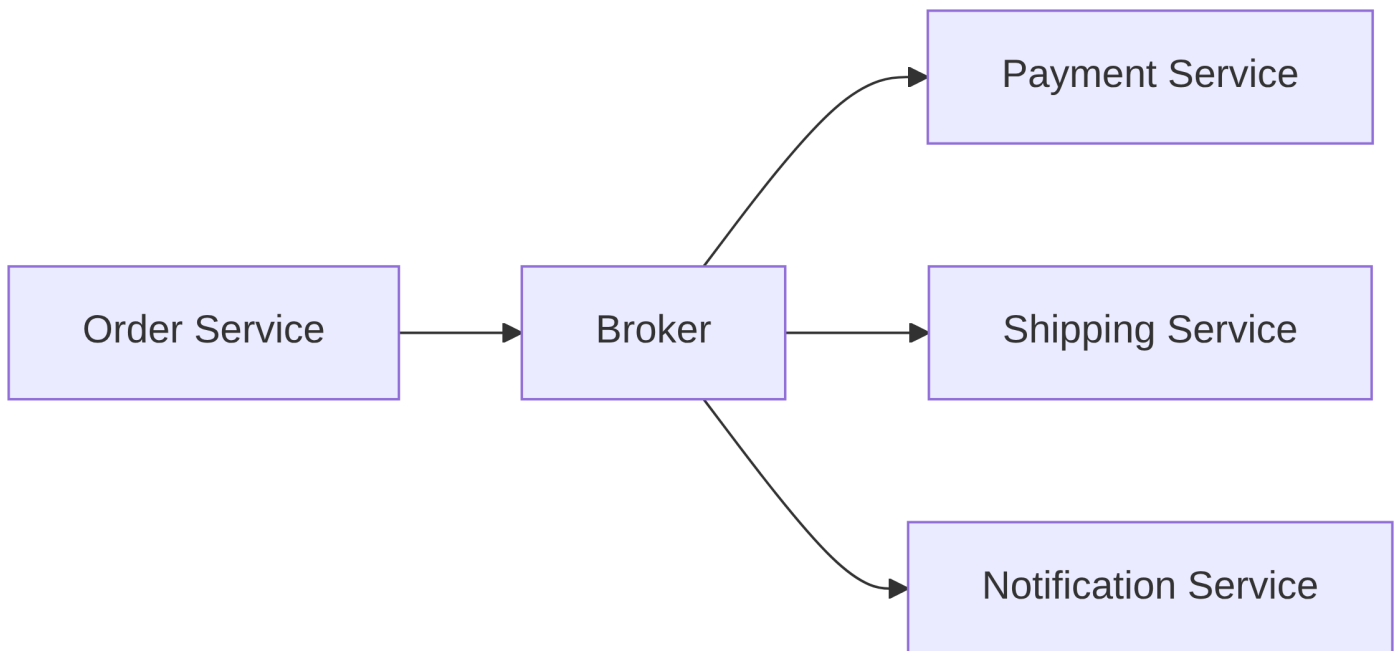
Key benefits:

- predictive maintenance
 - decoupled analytics
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Case Study 3: Microservices Event System

Scenario:

- Order service publishes events
- Payment, shipping, and notification services react



Key benefits:

- loose coupling
- scalability

When NOT to Use MQTT

Avoid MQTT for:

- large-scale streaming → use Kafka
- request/response APIs → use REST
- strong transactional workflows

Failure Scenarios

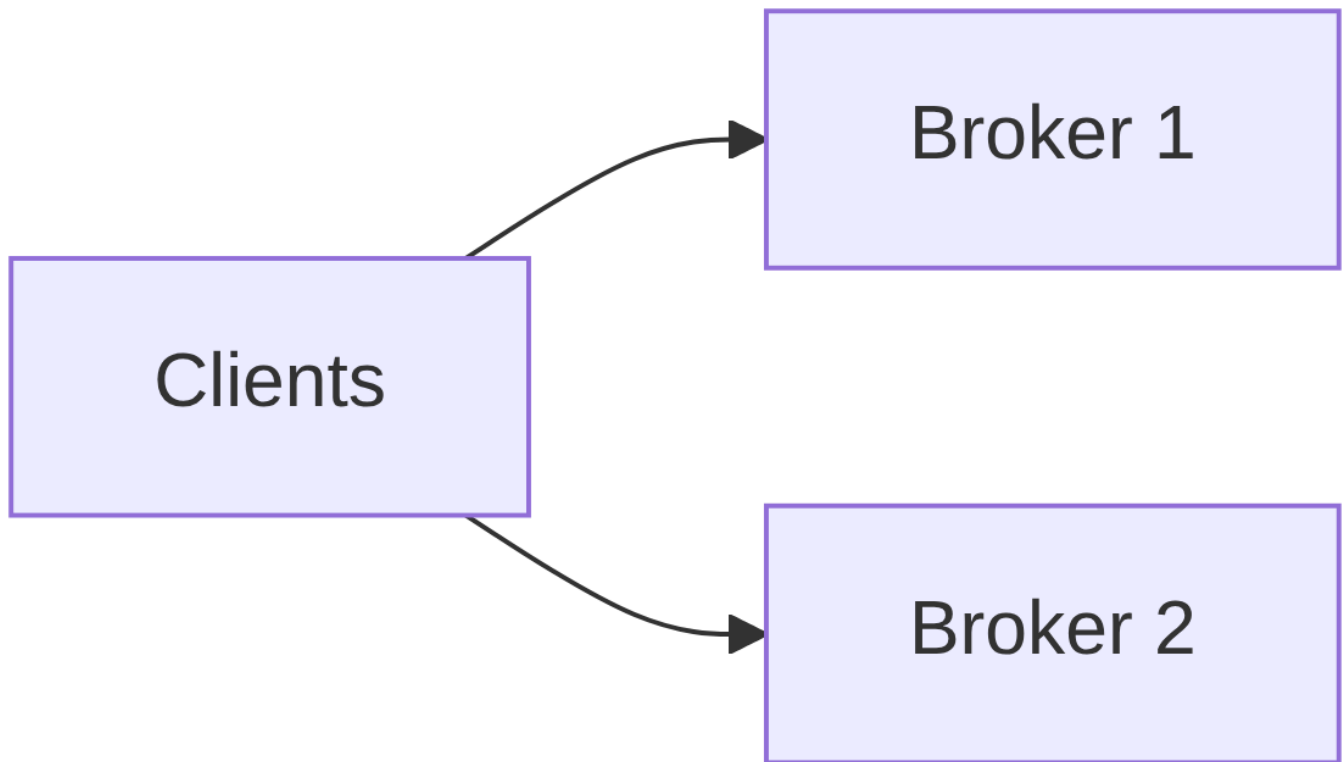
1. Broker Failure

Problem:

- Single broker becomes a bottleneck or point of failure

Solution:

- clustering (EMQX, HiveMQ)
- broker redundancy



2. Network Partition

Problem:

- Clients disconnect intermittently

Solutions:

- persistent sessions
- message buffering
- retries with backoff

3. Message Loss (QoS 0)

Problem:

- messages can be dropped

Solution:

- use QoS 1 or QoS 2
- implement retry logic

4. Duplicate Messages (QoS 1)

Problem:

- at-least-once delivery → duplicates

Solution:

- idempotent consumers
- deduplication

5. Slow Consumers

Problem:

- consumers cannot keep up

Solutions:

- shared subscriptions
 - horizontal scaling
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Resilience Patterns

Retry with Backoff

- exponential retry strategy
 - avoids overload during failures
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Circuit Breaker

- stop sending requests to failing services
 - recover after cooldown
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Dead Letter Topic (DLT)

- failed messages redirected

sensors/temp → processing → DLT

Idempotent Consumer

- process each message once logically
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Resources

- [MQTT Official Specification](#)
- [Eclipse Mosquitto](#)
- [HiveMQ Documentation](#)
- [MQTT Essentials Series](#)